

Beckham Moreland

Oakland, CA | (501) 507-9160 | beckham.hayes@gmail.com | [linkedin.com/in/becksm](https://www.linkedin.com/in/becksm)

EDUCATION

San Diego State University, College of Engineering
B.S. in Aerospace Engineering

Expected May 2029
Cumulative GPA: 3.7/4.0

EXPERIENCE

Smart Mobility Intern, NASA via Analytical Mechanics Associates – Mountain View, CA

June 2026 – Present

- Evaluated six UAS concepts across cost, endurance, manufacturability, and helicopter-deployment constraints through five Air National Guard consultations, helping down-select a low-cost multirotor for search-and-rescue
- Advanced the selected concept to flight-ready hardware in four days and helped deliver three airframe iterations in three weeks, contributing revisions to mounts, body geometry, landing fins, and avionics packaging
- Fabricated and integrated three additively manufactured airframes with nine flight-critical components while evaluating hardware options and preparing NASA procurement approval packages under NDAA constraints
- Designed a removable nose-ballast fixture and executed 10+ structured flight and drop tests across four CG configurations, exposing PLA degradation and controller vibration that drove material and geometry changes
- Built a Python/PyULog application to correlate PX4 vibration, motion, and actuator data with hardware changes, enabling data-driven anomaly diagnosis and redesign across three prototype iterations

Test and Visualization Engineering Intern, MeshMap Labs Inc. – San Francisco, CA

May 2025 – Sep 2025

- Planned and led UAV and RTK field tests across four sites, varying altitude, image overlap, and capture parameters to validate repeatable data-acquisition workflows under changing field conditions
- Analyzed post-flight data to identify coverage and georeferencing anomalies, executed targeted recapture tests, and verified corrective actions through fused UAV-imagery and terrestrial-LiDAR reconstructions

Full Stack Developer Intern, Axyz Technologies – Oakland, CA

Jun 2023 – Jul 2025

- Developed Python/Django and JavaScript front/back-end application features from user requirements, implementing database-backed views and application workflows that were validated through testing, debugging, and feedback

SKILLS

Flight Test & Systems: Flight-test cards, Test-procedure development, Preflight verification, PX4 ULog/telemetry data acquisition, Anomaly investigation, GNSS/IMU, Hardware integration

Programming & Data: Python, MATLAB, PyULog, pandas, matplotlib, Git/GitHub, Microsoft Excel

Design & Hardware: SOLIDWORKS, additive manufacturing, soldering, wiring, electromechanical assembly

PROJECTS

Fault-Tolerant Navigation Bench, Personal Project

In-Progress

- Designed and documented a bench-scale GNSS/IMU test platform using an MCU, IMU, GNSS, and microSD logging to evaluate sensor behavior under nominal and faulted conditions
- Developed Python logging, fault-injection, and analysis tools to test GPS dropout, IMU bias and noise, frozen packets, and signal loss through measured detection and recovery behavior

Student UAV Systems Research, SDSU Undergraduate Research

In-Progress

- Developing a UAV systems platform integrating onboard sensors, telemetry links, and ground-station controls to enable real-time vehicle-state monitoring and subsystem verification
- Evaluating MATLAB-based control and state-estimation outputs through error analysis, telemetry testing, and technical documentation to support future hardware integration and flight validation